Critical Points, Extrema & Graphing ANSWER KEY

PROCEDURE

1. Critical points. Find $f'(x)$ and factor. Critical points are x-values in the domain where $f'(c) = 0$ or $f'(c)$ is undefined.

2. Second Derivative Test. $f''(c) > 0$: local min. $f''(c) < 0$: local max. $f''(c) = 0$ or undefined: inconclusive.

3. First Derivative Test. Check the sign of f' on each side of c. + to −: local max. − to +: local min. No change: neither.

4. Global extrema. On a closed interval, compare f at every critical point and both endpoints. On all real numbers, think about end behavior.

The interactive flow chart is in the Calculus I Supplemental Materials: Local & Global Extrema, Flow Charts.

PROBLEM 1 $f(x) = x^3 - 3x^2 - 9x + 5$ on the interval $[-2, 6]$

(a) Find $f'(x)$ and factor it completely.

$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) =$ **$3(x - 3)(x + 1)$**

(b) Find the critical points.

f' is defined everywhere. $f'(x) = 0$ when $x = 3$ or $x = -1$. Both are in $[-2, 6]$.

Critical points: $x = -1$ and $x = 3$

(c) Find $f''(x)$, then use the **Second Derivative Test** on each critical point.

$f''(x) =$ 6x − 6

critical point c	f″(c)	f″(c) positive or negative?	local max, local min, or inconclusive
−1	−12	negative	local max
3	12	positive	local min

(d) Write each local extremum as a point (x, y).

Local max: (−1, 10) Local min: (3, −22)

(e) Find the **global** maximum and minimum of f on $[-2, 6]$. Fill in the table with every candidate.

x	−2	−1	3	6	
f(x)	3	10	−22	59	

Global max: 59 at x = 6 (endpoint) Global min: −22 at x = 3

Problem 2: When the Second Derivative Test Fails ANSWER KEY

PROBLEM 2 $f(x) = x^6 - 6x^4$ on all real numbers

(a) Find $f'(x)$ and factor it completely. (Hint: pull out the greatest common factor first, then look for a difference of squares.)

$f'(x) = 6x^5 - 24x^3 = 6x^3(x^2 - 4) =$ **$6x^3(x - 2)(x + 2)$**

(b) Find the critical points.

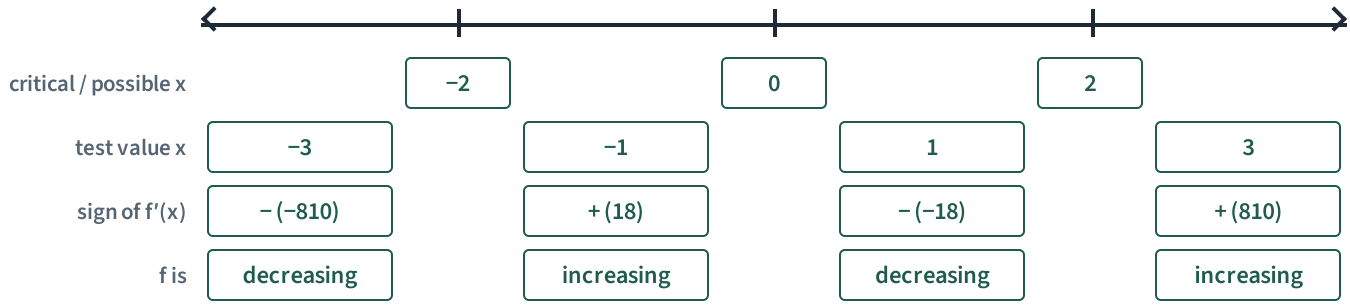
$f'(x) = 0$ when $x = 0$, $x = 2$, or $x = -2$. f' is defined everywhere.
Critical points: $x = -2, 0, 2$

(c) Find $f''(x)$, then use the **Second Derivative Test**. Write INCONCLUSIVE if it cannot decide.

$f''(x) =$ ~~$30x^4 - 72x^2 = 6x^2(5x^2 - 12)$~~

critical point c	$f''(c)$	result of the Second Derivative Test
-2	192	positive: local min
0	0	INCONCLUSIVE
2	192	positive: local min

(d) Use the **First Derivative Test** to settle every critical point. Complete the sign chart for $f'(x)$.



(e) Classify each critical point and give its coordinates.

$x = -2$: ~~local min at $(-2, -32)$~~

$x = 0$: ~~local max at $(0, 0)$~~

$x = 2$: ~~local min at $(2, -32)$~~

(f) Does f have a global maximum? A global minimum? Explain using end behavior and your values from (e).

No global maximum: $f(x) = x^4(x^2 - 6) \to \infty$ as $x \to \pm\infty$, so the graph rises without bound.
Global minimum: −32, at $x = -2$ *and* $x = 2$. The graph goes up on both ends, so the lowest point must be a critical point: f values are -32, 0, -32. The local max $(0, 0)$ is not the lowest.

Problem 3: A Cusp ANSWER KEY

PROBLEM 3 $f(x) = x^{2/3}(x - 5)$ on the interval $[-1, 8]$

(a) Rewrite f as a sum, $f(x) = x^{5/3} - 5x^{2/3}$. Then find $f'(x)$ and write it as a single fraction with the common factor pulled out.

$$f'(x) = \frac{5}{3}x^{2/3} - \frac{10}{3}x^{-1/3} = \frac{5}{3}x^{-1/3}(x - 2)$$
$$= \frac{\mathbf{5(x - 2)}}{\mathbf{3x^{1/3}}}$$

(b) Find **all** critical points in $[-1, 8]$. (Check where $f'(x) = 0$ *and* where $f'(x)$ is undefined but f is defined.)

$f'(x) = 0$ when $x - 2 = 0$, so $x = 2$.
$f'(x)$ is undefined when $x = 0$, but $f(0) = 0$ is defined, so $x = 0$ is also a critical point.
Critical points: x = 0 and x = 2

(c) Here is $f''(x) = \dfrac{10(x + 1)}{9x^{4/3}}$. Can the **Second Derivative Test** be used at each critical point? What does it say?

critical point c	f″(c)	what the Second Derivative Test says
0	undefined	cannot be used
2	≈ 1.32 (positive)	local min

(d) Use the **First Derivative Test** to classify every critical point. Complete the sign chart for $f'(x)$.



		0		2	
critical / possible x					
test value x	−1		1		3
sign of f′(x)	+ (5)		− (−5/3)		+ (≈ 1.16)
f is	increasing		decreasing		increasing

(e) Give the coordinates of each local extremum. Round decimals to two places.

x = 0: ~~local max at (0, 0)~~

x = 2: ~~local min at $(2, -3\cdot 4^{1/3}) \approx (2, -4.76)$~~

(f) Find the global maximum and minimum of f on $[-1, 8]$. Fill in every candidate. Is the local minimum also the global minimum?

x	−1	0	2	8	
f(x)	−6	0	≈ −4.76	12	

Global max: 12 at x = 8 Global min: −6 at x = −1

No. The local min at x = 2 has $f \approx -4.76$, but the endpoint $x = -1$ gives $f(-1) = (-1)^{2/3}(-6) = -6$, which is lower. The global minimum is at an endpoint.

Problem 4: Putting It All Together ANSWER KEY

PROBLEM 4 $f(x) = 3x^5 - 5x^3$ on all real numbers

You will use every tool from Problems 1 to 3, and also concavity. This page is the analysis. Page 5 is the graph.

(a) Critical points. Find $f'(x)$, factor completely, and list the critical points.

$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1) =$ **$15x^2(x - 1)(x + 1)$**

Critical points: $x = -1, 0, 1$

(b) Increasing and decreasing. Complete the sign chart for $f'(x)$.

		−1		0		1	
critical / possible x							
test value x	−2		−0.5		0.5		2
sign of $f'(x)$	+ (180)		− (−2.8)		− (−2.8)		+ (180)
f is	increasing		decreasing		decreasing		increasing

(c) Local extrema. Classify each critical point. $f''(x) = 60x^3 - 30x$. Use it to check −1 and 1, and explain what happens at 0.

$f''(-1) = -30 < 0$: **local max** at $(-1, 2)$. $f''(1) = 30 > 0$: **local min** at $(1, -2)$.

$f''(0) = 0$: the Second Derivative Test is inconclusive. In the sign chart, f' is negative on both sides of 0, so there is **no sign change: neither a max nor a min**. The graph flattens at $(0, 0)$ and keeps falling.

(d) Concavity and inflection points. Factor $f''(x) = 60x^3 - 30x$, find where it equals 0, and complete the sign chart for $f''(x)$.

$f''(x) =$ 30x(2x² − 1) zeros: $x =$ 0, ±√2/2 ≈ ±0.71

		−0.71		0		0.71	
critical / possible x							
test value x	−1		−0.5		0.5		1
sign of $f''(x)$	− (−30)		+ (7.5)		− (−7.5)		+ (30)
f is	concave down		concave up		concave down		concave up

(e) List the inflection points as coordinates (exact and decimal).

f'' changes sign at all three zeros, so all three are inflection points.

$(-\sqrt{2}/2, 7\sqrt{2}/8) \approx$ **$(-0.71, 1.24)$** **$(0, 0)$** $(\sqrt{2}/2, -7\sqrt{2}/8) \approx$ **$(0.71, -1.24)$**

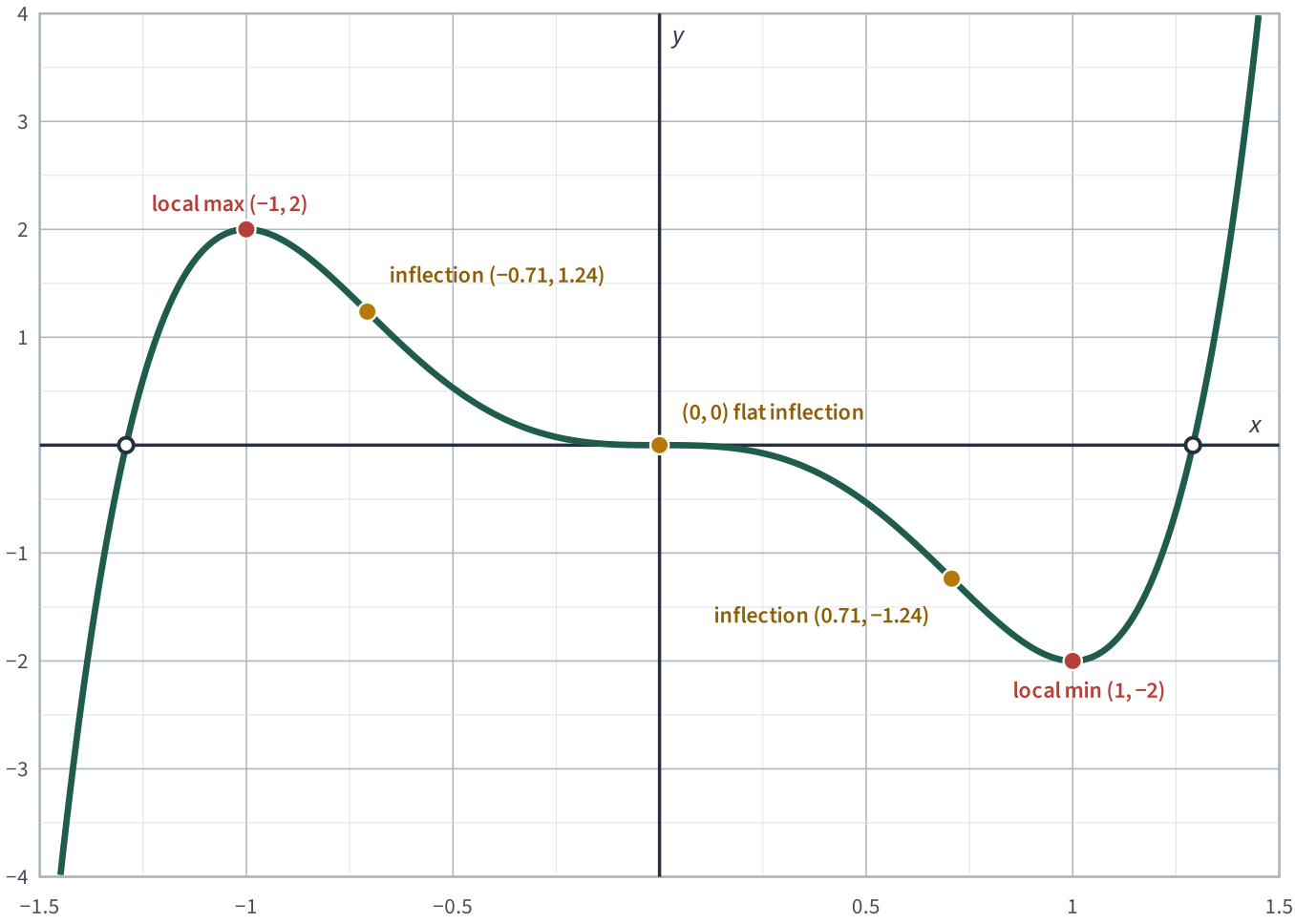
(f) Intercepts and global extrema. Find the y-intercept and any x-intercepts (factor f). Does f have global extrema? Why?

y-intercept: $f(0) = 0$, so **$(0, 0)$**. $f(x) = x^3(3x^2 - 5) = 0$ at $x = 0$ and $x = \pm\sqrt{(5/3)} \approx$ **± 1.29**.

No global extrema. f is odd degree: $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$. The local max 2 and local min −2 are not the highest or lowest values.

Problem 4, Continued: The Graph ANSWER KEY

Graph $f(x) = 3x^5 - 5x^3$ using your work from page 4. Plot and label every local extremum, inflection point, and intercept. Show concavity. The graph leaves the window near the top and bottom, so draw arrowheads where it goes off.



Key points. List them from left to right.

point (x, y)	what it is	point (x, y)	what it is
(−1.29, 0)	x-intercept	(0.71, −1.24)	inflection
(−1, 2)	local max	(1, −2)	local min
(−0.71, 1.24)	inflection	(1.29, 0)	x-intercept
(0, 0)	inflection, y-int		

Reflect. At x = 0 the Second Derivative Test told you nothing. In one or two sentences, what did the First Derivative Test tell you, and what does that look like on your graph?

The First Derivative Test showed f′ is negative on both sides of x = 0, so the sign does not change. That means (0, 0) is not a local max or min. On the graph the curve flattens to a horizontal tangent at the origin, then keeps falling. It is an inflection point where the concavity switches from up to down.